

Project Specification: Comparative Analysis of Image Compression Techniques

Course: CSE 4106 - Image Processing (CO2, CO3 Focus)

Project Title: Investigation and Performance Evaluation of Lossless and Lossy Image Compression Algorithms

Total Marks: 25 + 10 (Presentation/Demonstration + Report)

Due Dates:

- **Code & Report Submission:** [TBA]
- **Project Presentation:**[TBA]

1. Project Objective (Assessing K1, K2, P1, P3)

This project aims for a deep investigation and comparative analysis of various image compression techniques. Students will:

- **Identify** the fundamental principles and trade-offs inherent in image compression (lossless vs. lossy).
- **Select, implement, and evaluate** at least two distinct compression algorithms, covering both lossless and potentially a simplified lossy approach.
- **Utilize modern image processing tools and environments** for implementation and analysis.
- **Critically assess** the performance of these techniques based on compression ratio, reconstruction quality, and computational efficiency, and their impact on data storage and transmission.

Mapping:

- **K1, K2 (Problem Solving, Theoretical Foundation):** Students apply core image processing knowledge to a practical problem (data reduction).
- **P1 (Broadly-defined Engineering Problems):** Addressing real-world constraints of storage and bandwidth.
- **P3 (Structured Experimentation):** Systematic implementation and testing across different images.

2. Scope & Methodology

Students will choose at least **two distinct image compression algorithms** from the following list (or closely related variants covered in class), ensuring a diverse set for comparison:

- **Lossless Techniques:**
 - Run-Length Encoding (RLE)
 - Huffman Coding
 - Arithmetic Coding
 - LZW Encoding
- **Simplified Lossy Technique (Optional, for advanced comparison):**
 - Quantization-based approach (e.g., simple block-based transform followed by scalar quantization, or color quantization for palette images).
 - JPEG Compression Technique

For their chosen algorithms, they will:

1. **Implement** both the encoding and decoding processes.
2. **Acquire at least two diverse grayscale and/or color images** (e.g., one with large uniform areas, one with high detail/texture) for testing.
3. **Perform compression and decompression** using their implemented algorithms.
4. **Evaluate** the compressed images quantitatively (compression ratio, PSNR/SSIM for lossy) and qualitatively (visual inspection, fidelity).
5. **Compare** the selected algorithms against each other and potentially against a simple baseline (e.g., uncompressed size).

3. Deliverables and PO Mapping

The project will be assessed through the following deliverables, explicitly designed to map to the target POs, Ks, Ps, and As.

3.1. Project Report (Primary for PO5, Supports K1-K4, K6, K8, P1, P3, P5, A1, A3)

The report should be a comprehensive document (e.g., 10-15 pages, excluding appendices) that acts as the core technical documentation.

1. Introduction & Problem Statement (K1, K2, P1):

- Clearly define the image compression problem and its significance (storage, bandwidth limitations).
- Distinguish between lossless and lossy compression.
- State the project's specific objectives (evaluating techniques, trade-offs).

2. Theoretical Background & Mathematical Models (K2, K3, K4, P1):

- For **each** chosen compression algorithm:
 - Explain its underlying principles, mathematical/algorithmic steps, and data structures involved (e.g., Huffman tree, dictionary for LZW, run definition for RLE, arithmetic coding interval).
 - Discuss the theoretical basis for its compression efficiency.
 - Reference relevant literature (CO2, CO3 statements on evaluating techniques and mathematical models).

3. Methodology & Implementation Details (K3, K4, K6, P3, PO5):

- **Modern Tools Usage (PO5):** Clearly state the programming language (e.g., Python, MATLAB), specific libraries/frameworks used (e.g., NumPy, Pillow/PIL, collections for data structures), and justification for their selection.
- **Algorithm Steps:** Detail the implementation steps for both encoding and decoding for each algorithm.
- **Test Images:** Describe the chosen test images and justify why they are suitable for evaluating different compression scenarios (e.g., uniform vs. detailed).

4. Experimental Results & Evaluation (K3, K6, K8, P3, P5, A3):

- **Visual Results:** Present the original and reconstructed images (for lossy compression). For lossless, a visual comparison might be omitted unless artifacts are present from a baseline comparison. Use clear figures and captions.
- **Quantitative Metrics (K8, P5):**
 - **Compression Ratio (CR):** Calculate and present CR for all algorithms.
 - **Bits Per Pixel (BPP):** Calculate and present BPP.
 - **Reconstruction Quality (for Lossy only):** Use PSNR and SSIM to evaluate fidelity.
 - **Computational Efficiency:** Provide rough timings for encoding and decoding (optional but encouraged for P5).
- **Analysis of Results:** Discuss how the compression ratios vary with image content and algorithm choice. For lossy, analyze the trade-off between CR and reconstruction quality.

• 5. Discussion & Comparison (K6, K8, P5, A3):

- **Comparative Analysis:** Critically compare the performance (CR, BPP, quality, speed) of the implemented algorithms against each other and against uncompressed data.
- **Strengths & Weaknesses:** Detail the contexts where each algorithm excels or fails.
- **Trade-offs (P5):** Discuss the inherent trade-offs between compression ratio, reconstruction quality (for lossy), and computational complexity for each method.
- **Real-world Implications:** Conclude on the practical applicability of each technique for different image types or application requirements (e.g., archival, real-time streaming).

• 6. Conclusion & Future Work (K8, P5): Summarize key findings and suggest potential improvements, hybrid approaches, or future research directions.

• 7. References (K1): Proper academic citation of all sources.

• Appendices: Full source code with comments.

3.2. Project Presentation (Primary for PO4, Supports K1-K4, K6, K8, P1, P3, P5, A1, A2, A4)

The presentation (12-15 minutes, followed by 3-5 minutes Q&A) will serve as the primary assessment for **PO4 (Investigation)**. Students must demonstrate their investigative process.

- 1. Introduction & Problem (K1, P1, A1):** Clearly state the problem of image compression and its importance.
- 2. Compression Principles & Algorithms Overview (K2, K3, A1):** Briefly explain the core idea behind lossless vs. lossy compression and introduce the chosen algorithms.
- 3. Investigative Journey & Algorithm Details (PO4, K4, K6, P3, A2):**
 - This section is crucial for **PO4**. Students must articulate *how* they investigated the problem.
 - **Show the process:** Discuss why certain algorithms were chosen for comparison, what implementation challenges were encountered (e.g., managing symbol frequencies for Huffman/Arithmetic, dictionary growth for LZW), and how these were resolved.
 - **Deeper Dive:** For each implemented algorithm, present its core idea, key steps, and implementation nuances concisely.
- 4. Demonstration & Results (K3, K8, P5, A2):**
 - Present the original, encoded file size, and reconstructed images (for lossy).
 - Clearly display the calculated Compression Ratios, BPP, and (for lossy) PSNR/SSIM.
 - Highlight visual changes or artifacts from lossy compression.
- 5. Critical Analysis & Discussion (PO4, K6, K8, P5, A4):**
 - This is another key area for **PO4**. Engage in critical thinking.
 - **Compare and Contrast:** Directly compare the performance metrics and visual outcomes of the implemented algorithms.
 - **Trade-off Analysis:** Discuss the practical implications of the lossless/lossy trade-off for different applications.
 - **Limitations & Future Directions:** Acknowledge the limitations of the chosen methods and propose how they could be improved or what further investigations could be done (e.g., hybrid approaches, more advanced lossy methods like JPEG).
- 6. Q&A (PO4, A4):** Engage in a detailed Q&A session, demonstrating comprehensive understanding and ability to defend findings and discuss alternatives.

4. K, P, A Mapping Justification

- **Knowledge Profile (K):**
 - **K1 (Math, Science):** Applied in understanding information theory, probability for Huffman/Arithmetic coding.
 - **K2 (Discipline-specific):** Core image processing concepts of redundancy, entropy, and various compression schemes.
 - **K3 (Problem-Solving):** Selecting and implementing appropriate algorithms to reduce data size.
 - **K4 (Engineering Sciences):** Understanding the system perspective of data storage and transmission constraints.
 - **K6 (Design/Development):** Designing the experimental setup and implementation of encoders/decoders.
 - **K8 (Contemporary Issues):** Understanding the role of compression in modern digital media, the internet, and mobile applications.
- **Problem Level (P):**
 - **P1 (Broadly-defined):** Understanding the real-world need for efficient data storage and transmission.
 - **P3 (Structured Experimentation):** Designing, conducting, and analyzing controlled experiments with different compression algorithms and image types.
 - **P5 (Judgment & Evaluation):** Critically evaluating compression performance (CR, BPP, PSNR/SSIM) and drawing conclusions on suitability.
- **Assessment (A):**
 - **A1 (Proposal/Plan):** Implicit in the project setup and initial understanding.
 - **A2 (Demo/Application):** Through implementation and showing compressed/decompressed images and metrics.
 - **A3 (Report):** Comprehensive written documentation of the project.
 - **A4 (Presentation):** Oral communication of findings, defending choices, and engaging in discussion.

5. Marking Scheme (25 + 10 Marks)

- **Problem Identification & Objective Definition (3 + 1 Mark):** Clarity, relevance, and scope (lossless vs. lossy).
- **Theoretical Background & Mathematical Models (4 + 2Marks):** Accuracy, depth of explanation for chosen algorithms (e.g., Huffman tree, LZW dictionary).
- **Implementation (6 + 2 Marks):** Correctness, efficiency, and clarity of code; effective use of modern tools (PO5).
- **Experimental Results & Evaluation (6 + 2 Marks):** Presentation of visual and quantitative results (CR, BPP, PSNR/SSIM); insightful analysis of compression efficiency and quality (CO2, CO3).
- **Discussion & Comparison (4 + 2 Marks):** Critical comparison of algorithms; analysis of strengths/weaknesses and suitability; demonstration of investigative depth (PO4).
- **Report & Presentation (2 + 1 Marks):** Overall clarity, structure, visuals, and oral delivery; effective communication of investigative process (PO4) and tool usage (PO5).